

Problem Set 13

Problem 12.2.3: $x = \rho \cos \phi$ and $y = \rho \sin \phi$ imply

$$\begin{aligned}\partial_\rho &= \frac{\partial x}{\partial \rho} \partial_x + \frac{\partial y}{\partial \rho} \partial_y = \cos \phi \partial_x + \sin \phi \partial_y, \\ \partial_\phi &= \frac{\partial x}{\partial \phi} \partial_x + \frac{\partial y}{\partial \phi} \partial_y = -\rho \sin \phi \partial_x + \rho \cos \phi \partial_y.\end{aligned}$$

Invert this to get

$$\begin{aligned}\partial_x &= \cos \phi \partial_\rho - \frac{1}{\rho} \sin \phi \partial_\phi, \\ \partial_y &= \sin \phi \partial_\rho + \frac{1}{\rho} \cos \phi \partial_\phi.\end{aligned}$$

Therefore, since $L_z = XP_y - YP_x$, acting on wavefunctions in the position representation, $L_z = -i\hbar(x\partial_y - y\partial_x) = -i\hbar[\rho \cos \phi(\sin \phi \partial_\rho + \frac{1}{\rho} \cos \phi \partial_\phi) - \rho \sin \phi(\cos \phi \partial_\rho - \frac{1}{\rho} \sin \phi \partial_\phi)] = -i\hbar(\cos^2 \phi + \sin^2 \phi) \partial_\phi = -i\hbar \partial_\phi$.

Alternatively, we can deduce this by the action of rotations on wavefunctions. On the one hand,

$$U(\epsilon \hat{\mathbf{k}}) [\psi(\rho, \phi)] = e^{-i\epsilon L_z/\hbar} [\psi(\rho, \phi)] = \psi(\rho, \phi) - \frac{i\epsilon}{\hbar} L_z [\psi(\rho, \phi)] + \mathcal{O}(\epsilon^2),$$

and on the other hand,

$$U(\epsilon \hat{\mathbf{k}}) [\psi(\rho, \phi)] = \psi(\rho, \phi - \epsilon) = \psi(\rho, \phi) - \epsilon \frac{\partial \psi}{\partial \phi}(\rho, \phi) + \mathcal{O}(\epsilon^2).$$

Comparing at order ϵ gives $-(\partial \psi / \partial \phi) = -(i/\hbar) L_z [\psi]$, or $L_z = -i\hbar \partial_\phi$.

Problem 12.3.3: The state $|\psi\rangle$ in question factorizes as the tensor product of a radial and an angular part, $|\psi\rangle = |A\rangle|B\rangle$, so that $\psi(\rho, \phi) = \langle \rho, \phi | A, B \rangle = A(\rho)B(\phi)$. We saw in class that the L_z eigenfunctions, $\Phi_m(\phi)$, are functions of ϕ alone, so we can write

$$\psi(\rho, \phi) = A''(r) \cos^2 \phi = A'(r)(e^{2i\phi} + 2 + e^{-2i\phi}) = A(r)(\Phi_{+2}(\phi) + 2\Phi_0(\phi) + \Phi_{-2}(\phi)).$$

Whatever $A(r)$ is, ψ is normalized, so

$$\begin{aligned}1 &= \int_0^\infty \rho d\rho \int_0^{2\pi} d\phi |\psi|^2 = \left(\int_0^\infty \rho d\rho |A|^2 \right) \left(\int_0^{2\pi} d\phi |\Phi_{+2} + 2\Phi_0 + \Phi_{-2}|^2 \right) \\ &= \left(\int_0^\infty \rho d\rho |A|^2 \right) (1 + 4 + 1),\end{aligned}$$

since the Φ_m are normalized so that $\int_0^{2\pi} d\phi \Phi_m^* \Phi_n = \delta_{m,n}$. This thus implies that on the radial part of the Hilbert space

$$\langle A|A \rangle = \left(\int_0^\infty \rho d\rho |A|^2 \right) = \frac{1}{6}. \quad (1)$$

Now, to compute the probabilities $\text{Prob}(\ell_z = m\hbar)$, we need the norm-squared of the projection of $|\psi\rangle$ onto the $L_z = m\hbar$ eigenspace. This eigenspace is spanned by the orthonormal set $\{|n, \Phi_m\rangle\}$ where $|n\rangle$ is (any) complete orthonormal basis of radial states. We then have

$$\begin{aligned} \text{Prob}(\ell_z = m\hbar) &= \sum_n |\langle n, \Phi_m | A, B \rangle|^2 = \sum_n |\langle n | A \rangle \langle \Phi_m | B \rangle|^2 = |\langle \Phi_m | B \rangle|^2 \cdot \sum_n \langle A | n \rangle \langle n | A \rangle, \\ &= |\langle \Phi_m | B \rangle|^2 \cdot \langle A | A \rangle = \frac{1}{6} |\langle \Phi_m | B \rangle|^2 = \frac{1}{6} \left| \int_0^{2\pi} d\phi \Phi_m^*(\phi) B(\phi) \right|^2, \end{aligned}$$

where in the second line I used the completeness of the radial basis, and the result (1) from above. Using our result that $B(\phi) = \Phi_{+2}(\phi) + 2\Phi_0(\phi) + \Phi_{-2}(\phi)$ and the orthonormality of the Φ_m , we then get

$$\begin{aligned} \text{Prob}(\ell_z = 2\hbar) &= \frac{1}{6} \left| \int_0^{2\pi} d\phi \Phi_2^*(\Phi_2 + 2\Phi_0 + \Phi_{-2}) \right|^2 = \frac{1}{6}, \\ \text{Prob}(\ell_z = 0) &= \frac{1}{6} \left| \int_0^{2\pi} d\phi \Phi_0^*(\Phi_2 + 2\Phi_0 + \Phi_{-2}) \right|^2 = \frac{2}{3}, \\ \text{Prob}(\ell_z = -2\hbar) &= \frac{1}{6} \left| \int_0^{2\pi} d\phi \Phi_{-2}^*(\Phi_2 + 2\Phi_0 + \Phi_{-2}) \right|^2 = \frac{1}{6}. \end{aligned}$$

Problem 12.3.7: In the coordinate basis in polar coordinates

$$\begin{aligned} H &\rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 + \frac{\mu\omega^2}{2} \rho^2, \quad \text{with} \quad \nabla^2 = \partial_\rho^2 + \frac{1}{\rho} \partial_\rho + \frac{1}{\rho^2} \partial_\phi^2, \\ L_z &\rightarrow -i\hbar \partial_\phi. \end{aligned}$$

- (1) From this it is obvious that $[H, L_z] = 0$ since $[X, \partial_\phi] = 0$ if X doesn't depend explicitly on the ϕ coordinate. So choose as the H eigenstates $|E, m\rangle \sim R_E(\rho) \Phi_m(\phi)$. Since $-i\hbar \partial_\phi \Phi_m = L_z |\Phi_m\rangle = m\hbar |\Phi_m\rangle = m\hbar \Phi_m$, we have $\partial_\phi \Phi_m = im \Phi_m$. So $H|E, m\rangle = E|E, m\rangle$ reads in the polar coordinate basis

$$\left(-\frac{\hbar^2}{2\mu} \left(\partial_\rho^2 + \frac{1}{\rho} \partial_\rho + \frac{1}{\rho^2} \partial_\phi^2 \right) + \frac{\mu\omega^2}{2} \rho^2 - E \right) (R_E(\rho) \Phi_m(\phi)) = 0,$$

which implies

$$\left(-\frac{\hbar^2}{2\mu} \left(\partial_\rho^2 + \frac{1}{\rho} \partial_\rho - \frac{m^2}{\rho^2} \right) + \frac{\mu\omega^2}{2} \rho^2 - E \right) R_E(\rho) = 0. \quad (2)$$

- (2) Assume $R_E(\rho) \sim \rho^\alpha + \mathcal{O}(\rho^{\alpha+1})$ as $\rho \rightarrow 0$. Then $\partial_\rho R_E \sim \alpha \rho^{\alpha-1} + \mathcal{O}(\rho^\alpha)$ and $\partial_\rho^2 R_E \sim \alpha(\alpha-1)\rho^{\alpha-2} + \mathcal{O}(\rho^{\alpha-1})$, so the leading terms of equation (2) are

$$-\frac{\hbar^2}{2\mu} (\alpha(\alpha-1) + \alpha - m^2) \rho^{\alpha-2} + \frac{\mu\omega^2}{2} \rho^{\alpha+2} - E\rho^\alpha + \mathcal{O}(\rho^{\alpha-1}) = 0.$$

So for the leading term to vanish as $\rho \rightarrow 0$ we must have $\alpha(\alpha-1) + \alpha - m^2 = 0$ which implies $\alpha^2 = m^2$, or $\alpha = |m|$. (Note that α must be non-negative for $R_E(\rho)$ to be finite as $\rho \rightarrow 0$.)

- (3) Assume $R_E(\rho) \sim \rho^n e^{-\beta\rho^\alpha}$ as $\rho \rightarrow \infty$ with both α and β positive (so that R_E goes to zero at infinity). Then, up to lower-order terms, $\partial_\rho R_E \sim -\alpha\beta\rho^{n+\alpha-1}e^{-\beta\rho^\alpha}$, and $\partial_\rho^2 R_E \sim \alpha^2\beta^2\rho^{n+2\alpha-2}e^{-\beta\rho^\alpha}$. Therefore, the leading terms in the large- ρ limit of eqn. (2) are

$$\left(-\frac{\hbar^2}{2\mu} (\alpha^2\beta^2\rho^{n+2\alpha-2} - \alpha\beta\rho^{n+\alpha-2}) + \frac{\mu\omega^2}{2}\rho^{n+2} \right) e^{-\beta\rho^\alpha} = 0.$$

A cancellation can only occur (between the first and third terms) if $\alpha = 2$, implying

$$-\frac{\hbar^2}{2\mu}\beta^2 + \frac{\mu\omega^2}{2} = 0,$$

or, $\beta = \mu\omega/(2\hbar)$. This then implies that $R_E \sim e^{-\mu\omega\rho^2/(2\hbar)}$ as $\rho \rightarrow \infty$.

- (4) Change variables from ρ to y and E to ε by

$$\rho \rightarrow \sqrt{\frac{\hbar}{\mu\omega}} y, \quad \partial_\rho \rightarrow \sqrt{\frac{\mu\omega}{\hbar}} \partial_y, \quad E \rightarrow \hbar\omega\varepsilon, \quad R_E(\rho) \rightarrow R_E(y).$$

Then eqn. (2) becomes

$$\left(\partial_y^2 + \frac{1}{y}\partial_y - \frac{m^2}{y^2} - y^2 + 2\varepsilon \right) R_E = 0. \quad (3)$$

- (5) Plug the *ansatz* $R_E = y^{|m|}e^{-y^2/2}U$ into eqn. (3). To do this, first compute

$$\begin{aligned} \partial_y R_E &= [(|m|y^{-1} - y)U + U'] y^{|m|}e^{-y^2/2}, \\ \partial_y^2 R_E &= [(|m|(|m|-1)y^{-2} - 2|m|-1+y^2)U + (2|m|y^{-1} - 2y)U' + U''] y^{|m|}e^{-y^2/2}. \end{aligned}$$

Then eqn. (3) becomes, after some cancellations and dividing through by $y^{|m|}e^{-y^2/2}$,

$$U'' + [(2|m|+1)y^{-1} - 2y]U' + [2\varepsilon - 2|m| - 2]U = 0. \quad (4)$$

(6 & 7) Plug $U = \sum_{r=0}^{\infty} c_r y^r$ into eqn. (4) to get

$$\begin{aligned}
0 &= \sum_{r=0}^{\infty} \{r(r-1)c_r y^{r-2} + (2|m|+1)rc_r y^{r-2} - 2rc_r y^r + (2\varepsilon - 2|m| - 2)c_r y^r\} \\
&= \sum_{r=0}^{\infty} rc_r [r + 2|m|] y^{r-2} + \sum_{r=0}^{\infty} 2c_r [\varepsilon - |m| - 1 - r] y^r \\
&= c_1(1 + |m|)y^{-1} + \sum_{r=0}^{\infty} \{(+2)c_{r+2}[r + 2 + 2|m|] + 2c_r[\varepsilon - |m| - 1 - r]\} y^r,
\end{aligned}$$

where I shifted $r \rightarrow r + 2$ in the first summation. Since each coefficient of a different power of y must vanish, we have

$$\begin{aligned}
0 &= c_1(1 + 2|m|), \\
c_{r+2} &= -\frac{2(\varepsilon - |m| - 1 - r)}{(r+2)(r+2+2|m|)} c_r, \quad r \in \{0, 1, 2, \dots\}.
\end{aligned} \tag{5}$$

If the series does not terminate, then for large r we have $c_{r+2} \approx (2/r)c_r$ which implies that $c_{2n} \sim 1/(n-1)!$. Therefore, $U \sim \sum_r c_r y^r \sim \sum_n y^{2n}/(n-1)! \sim y^2 e^{+y^2}$. So, for large y , $R_E \sim e^{-y^2/2} U \sim e^{-y^2/2} e^{+y^2} \sim e^{+y^2/2} \rightarrow \infty$, which violates the boundary conditions. Therefore the series must terminate for some r . This implies that there must exist an r such that

$$\frac{2(\varepsilon - |m| - 1 - r)}{(r+2)(r+2+2|m|)} = 0$$

which can occur whenever

$$\varepsilon = |m| + 1 + r, \quad r \in \{0, 1, 2, \dots\}.$$

But eqn. (5) implies $c_1 = 0$, and so all $c_{\text{odd}} = 0$. Therefore we can take r to be even: $r \equiv 2k$. Thus the final answer for the (dimensionless) energy eigenvalues is

$$\varepsilon = |m| + 1 + 2k, \quad k \in \{0, 1, 2, \dots\},$$

or, in terms of the original dimensionful energy units

$$E = \hbar\omega(|m| + 1 + 2k).$$

(8) Define $n \equiv 2k + |m|$. Then, for a given n , $|m| = n - 2k$, so $|m| \in \{n, n-2, n-4, \dots, (1 \text{ or } 0)\}$ where the last entry is 1 if n is odd, or 0 if n is even. Since m can have either sign, we thus get that

$$m \in \{n, n-2, n-4, \dots, -n+2, -n\}.$$

The number of entries in this list is $n + 1$, which is thus the degeneracy of the n -th energy level.

- (9) • For the $n = 0$ state we must have $m = k = 0$, implying $U = \text{constant}$. Therefore $R_0 = C''y^0e^{-y^2/2} = C'e^{-\mu\omega\rho^2/(2\hbar)}$ for some constant C' . Recalling that $\Phi_0 = (2\pi)^{-1/2}$ is itself just a constant, we get that the total wavefunction must be of the form $\psi_{0,0} = Ce^{-\mu\omega\rho^2/(2\hbar)}$. We can determine the constant by normalizing:

$$1 = \int_0^\infty \rho d\rho \int_0^{2\pi} d\phi |\psi_{0,0}|^2 = 2\pi|C|^2 \int_0^\infty \rho d\rho e^{-\mu\omega\rho^2/\hbar},$$

which implies that $C = \sqrt{\mu\omega/(\pi\hbar)}$ (up to a phase), so

$$\psi_{0,0} = \sqrt{\frac{\mu\omega}{\pi\hbar}} e^{-\mu\omega\rho^2/(2\hbar)}.$$

- For the $n = 1$ states we have $m = \pm 1$ and $k = 0$, which again imply that $U = \text{constant}$, and so $R_{1,\pm 1} = C''ye^{-y^2/2} = C'\rho e^{-\mu\omega\rho^2/(2\hbar)}$. Since $\Phi_{\pm 1} = (2\pi)^{-1/2}e^{\pm i\phi}$, we get $\psi_{1,\pm 1} = C\rho e^{-\mu\omega\rho^2/(2\hbar)}e^{\pm i\phi}$. Normalizing gives

$$1 = \int_0^\infty \rho d\rho \int_0^{2\pi} d\phi |\psi_{1,\pm 1}|^2 = 2\pi|C|^2 \int_0^\infty \rho^3 d\rho e^{-\mu\omega\rho^2/\hbar},$$

which implies that $C = \mu\omega/(\hbar\sqrt{\pi})$ (up to a phase), so

$$\psi_{1,\pm 1} = \frac{\mu\omega}{\sqrt{\pi\hbar}} \rho e^{-\mu\omega\rho^2/(2\hbar)} e^{\pm i\phi}.$$

- (10) • The $n = 0$ wavefunction *must* equal the one in Cartesian coordinates since it has degeneracy 1, *i.e.*, it is unique.
• The two $n = 1$ solutions

$$\psi_{1,\pm 1} = C(\rho \cos \phi \pm i\rho \sin \phi) e^{-\mu\omega\rho^2/(2\hbar)} = C(x \pm iy) e^{-\mu\omega(x^2+y^2)/(2\hbar)}$$

are linear combinations of the two cartesian coordinate solutions $xe^{-\mu\omega(x^2+y^2)/(2\hbar)}$ and $ye^{-\mu\omega(x^2+y^2)/(2\hbar)}$.

- Parity acts on the cartesian coordinates as $\Pi : (x, y) \rightarrow (-x, -y)$, which translates in polar coordinates to $\Pi : (\rho, \phi) \rightarrow (\rho, \phi + \pi)$. Thus $\Pi : \psi_{n,m} \sim f(\rho)e^{-im\phi} \rightarrow (-)^m \psi_{n,m}$. But n is even/odd if m is even/odd, so $(-)^m \psi_{n,m} = (-)^n \psi_{n,m}$.